

Optimizing Stroke Detection: An Analysis of Different Feature Selection Approaches

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Abstract

Stroke is a serious condition that can make it impossible for some body parts to move. When blood flow to a portion of the brain is suddenly cut off, a stroke results. A stroke patient needs to receive treatment right away because delaying care could seriously harm their health. This research aims to identify the optimal model with important features based on several machine learning techniques that can predict the risk of stroke from the provided data. The combined dataset of publicly available data with 733 collected data is used to train the model. We have analyzed the result of 4 different feature selection techniques - Information gain, Lasso L1, Fisher score and Kendall's Tau, to reduce dimensionality and ensure a cost-effective and scalable stroke detection system. Also, 5-fold cross-validation with Grid Search CV hyperparameter tuning is used to remove any potential overfitting issues. In addition, we have evaluated the model using the Brier score, G-mean, Matthews Correlation Coefficient (MCC), and H-measure. Lasso L1 emerges as the most compelling feature selection technique for this project. With this technique, the best accuracy of 90%, H-measure of 89.9% and AUC score of 96% are provided by Gradient Boosting among the eight machine learning algorithms we have used.

CCS Concepts

• Computing methodologies • Machine learning • Machine learning algorithms • Feature selection

Keywords: Stroke; Machine Learning; Lasso L1; Fisher score; Kendall's Tau; Information gain; Matthews Correlation Coefficient; Brier score; AUC Score.

ACM Reference format:

Monirul Islam Mahmud, Md Shihab, Reza and Syeda Shabnam Khan. 2024. Optimizing Stroke Detection: An Analysis of Different Feature Selection Approaches. In *Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '24)*, October

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UbiComp Companion '24, October 5–9, 2024, Melbourne, VIC, Australia

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ACM ISBN 979-8-4007-1058-2/24/10. <https://doi.org/10.1145/3675094.3677602>

5–9, 2024, Melbourne, VIC, Australia. ACM, New York, NY, USA, 4 pages.
<https://doi.org/10.1145/3675094.3677602>

1 Introduction

The updated ACM template, user manuals, samples, and required fonts, all are available at the URL. A stroke is a serious medical condition that arises when there is reduced or blocked blood circulation to a certain area of the brain. Thus, brain tissue cannot receive oxygen and nutrients, which results in an ischemic stroke. The World Stroke Organization estimates that 13 million people worldwide get a stroke each year, with 5.5 million of them dying as a result [1]. Stroke does not only affect elderly people but is also the leading cause of death and disability in the world [2]. The American Heart Association states that because stroke has a high mortality rate, it is regarded as a serious health problem [3]. While traditional statistical methods have long been employed for stroke prediction, recent advances in computing and global IT transformation have significantly enhanced these approaches, making machine learning (ML) and deep learning (DL) techniques increasingly practical and effective for stroke prediction. M. C. Das et al. [18] used machine learning algorithms for Stroke Prediction and got the best result given by Random Forest. Different Machine Learning models are implemented and ended up getting the best result given by Random Forest and Decision Tree in the paper of A. Patel [19]. Various feature selection techniques have been employed in stroke prediction research, including optimized feature selection, PCA-FA, FPCA, wrapper-based methods, and XGBoost, in these research [21], [22], [23], [24], and [25]. From these reviews, we found the majority of existing research heavily relied on public datasets only. There were lacking performance and different feature selections to observe the crucial features of stroke.

This study benefits the Ubicomp/ISWC community by offering significant new information about how stroke prediction can easily be detectable with important features. So, we will conduct this research using a mixture of publicly available and collected datasets from 2 Bangladeshi hospitals while analyzing four different feature selections to find out which features are more salient for detecting stroke. Thus far, there has been limited research comparing the results and key factors of stroke using different feature selection techniques such as Fisher score, Kendall's Tau, Lasso L1 and Information Gain in a single paper. The result and scalability of stroke prediction can be enhanced by using feature selection techniques to reduce the dimensionality,

thus lowering the computational burden of models as well as mitigating multicollinearity.

2 Literature Review

In the paragraph below, existing pieces of literature were reviewed to gain essential information about several models, different types of feature scaling techniques, and sampling techniques that were used in stroke prediction.

Atul Kumar Uttam at [6] used Kaggle data where 11 separate variables are included in the dataset with 5110 records out of which 4861 records of no stroke and 249 records of stroke are present. R.K Kavitha at el. [7] used 5110 records from Kaggle with 10 features. Again, public data from Kaggle is used by Chetan Sharma et al. [8] also. Govindarajan at el. [9] used 507 patient data collected from Sugam Multispecialty Hospital, Kumbakonam. Elias et al. [11] used the Kaggle dataset, where they focused on participants who were over 18 years old. The number of rows were 3254, and 10 attributes. This research [7] uses feature selection techniques such as Principal Component Analysis (PCA), Chi-Square, PCA combined with Chi-Square, and Chi-Square with outlier removal. Ting Zuo and Fenglian Li et al. [21] proposed stroke classification model which had improvements in optimizing the feature selection, MRF based SMOTE oversampling, and Dueling DQN algorithm than the previous models. In similar research carried out by Saad Sahriar and Sanjida et al. [22], Akther used statistical tests to ascertain the risk factors relating to stroke. They further enhanced the efficiency of the used machine learning algorithms by applying three feature extraction techniques, namely, PCA, FA and FPCA leading to average accuracy of 92%. Thus, 55 % and an AUC of 98. 15% using the random forests model. Electronic health records were used by Santwana Gudadhe and Anuradha Thakare et al. [23] whereby they used the wrapper-based method to evaluate the features that would enhance the prediction of strokes accurately. Logistic Regression (LR), Random Forest (RF), XGBoost, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Multi-Layer Perceptron (MLP) were used by C. Kokkotis et al. [12]. The LR classifier achieved the best scores in accuracy. With a score of 90.36% in [14], Random Forest is the algorithm that performs the best in terms of accuracy. A total of ten classifiers were used in the proposed study [15] to predict strokes. The results showed that the weighted voting classifier outperformed the base classifiers with a 97% accuracy rate. Random Forest (RF), Support Vector Machine (SVM), and Decision Tree (DT) classifiers are applied in [16] and Random Forest gave the maximum accuracy of 95.30%. S. K. Satapathy, Thanka at el. [20] got their best result by ANN model with 99.3% accuracy by using RandomSearch hyperparameter tuning and also SMOTE & Random Undersampling technique.

3 Methodology

This section explains the proposed system's methodology as well as how it was successfully implemented. This part is divided into six subsets: (1) Project Workflow (2) Dataset Details (3) Data Preprocessing & Feature Extraction (4) Feature selection techniques (5) Models and Evaluation Criteria.

3.1 Project Workflow

After collecting our dataset from Kaggle [4], we merged 733 records from Bangladeshi hospitals nearby to strengthen our model. To achieve the optimum outcome, we deployed eight machine learning models after using four different feature selection techniques in order to remove variables without losing the pattern and analyze their results to separate observe the important features of stroke.

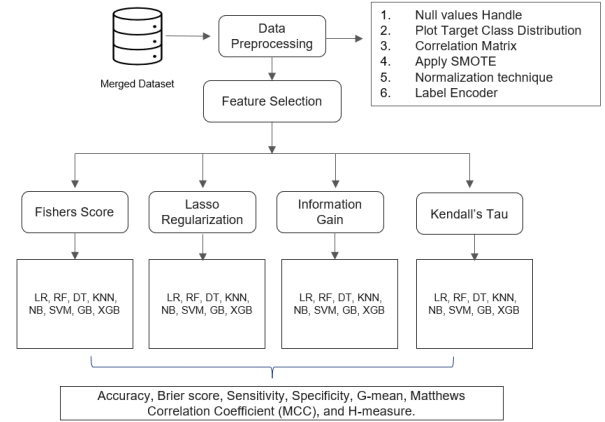


Figure 1: Project Workflow Diagram.

3.2 Dataset Details

We have used a merged dataset of public and collected stroke patient data. This dataset consists of 12 features with 5894 data, where 5111 data are from publicly available data and 733 collected data from 2 Bangladeshi hospitals. A consent form was given to all the patients where the purpose of the data collection was clearly mentioned, the data would only be used for research purposes and every consent form was signed by each patient. It should be noted that 516 patients from these 733 data were stroke patients and the rest of them were non-stroke.

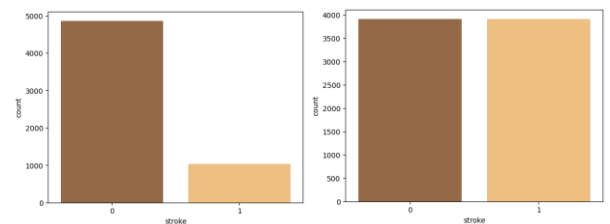


Figure 2: Distribution plot before and after using SMOTE.

3.3 Data Preprocessing & Feature Extraction

The dataset's null values were analyzed afterward. BMI and smoking status have been found to have 341 and 145 null values, respectively. Mean values were used to fill up the null values of BMI and the smoking_status null values were filled in with "unknown." We have used the Normalization technique so that all numerical values fall between the same range of 0 to 1 and similarly scale all the values. The label encoder was then used to encode all the categorical features. Categories each received a unique integer value in Label Encoding. Next, the dataset was

divided into a training set and a testing set, with the training dataset using 80% of the data and the testing dataset using 20%. Boxplot and pair plots were plotted to determine the relations between the features. Following the analysis of the count plot, it became noticeable that there were extremely few people with strokes, while on the other hand, most of the data were from non-stroke patients (Figure 2). So, we have used the SMOTE technique to balance the dataset.

3.4 Feature selection techniques

Information Gain, Fisher score, Lasso L1 and Kendall's Tau were implemented in this research study so far to analyze the outcomes of all techniques and choose one feature selection technique for stroke prediction. The decrease in entropy resulting from a dataset's transformation is measured as information gain [26]. The procedure involves figuring out each feature's information gain, which measures how much data a feature provides for classification. To accomplish shrinkage and variable selection at the same time, LASSO (Least Absolute Shrinkage and Selection Operator) is a helpful tool, as per Y. Kim [27]. Given that LASSO employs the L1 penalty, the optimization ought to depend on the computationally demanding quadratic program (QP) or general non-linear program. Selected features are shown with their coefficients in Figure 3.

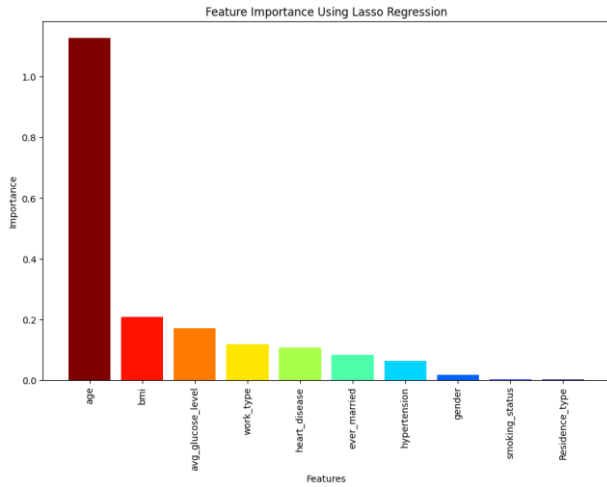


Figure 3: Features with Coefficient in Lasso L1

A non-parametric way to measure the relationships between ranked data columns is to use Kendall's Tau [28]. A relationship is perfect when the correlation coefficient is 1, and there is no relationship when it is 0. A feature selection method called Fisher's Discriminant Ratio or Fisher's Score [29] ranks feature according to how well they can distinguish between different classes in a dataset. The ratio of within-class to between-class variance results in the Fisher's Score. A trait with a higher Fisher's Score is considered more valuable and discriminative for classification.

3.5 Models and Evaluation Criteria

After splitting our overall dataset into two parts – Train Set (80%) and Test Set (20%), we have applied eight machine learning algorithms; they are Random Forest, Decision Tree, K Nearest

Neighbors, Naïve Bayes, SVM, Gradient Boosting, XGBoost and Logistic Regression. We use K-fold validation with a 5-fold train set where the train set is divided into 5 equal-sized folds or subsets. The model will train five times in total, using different test folds each time. Additionally, GridsearchCV is employed for hyperparameter tuning to prevent overfitting of the models. We have evaluated model performance by calculating accuracy, Brier score, Sensitivity, Specificity, G-mean, Matthews Correlation Coefficient (MCC), and H-measure.

4 Result Analysis & Discussion

In this section, the outcomes for the prediction performance of the proposed ML models are summarized. The performance of the models is evaluated based on accuracy, F1 score, brier score, G-mean, H-measure and MCC. Firstly, we have applied all models without using any feature selection techniques, where we have found Gradient Boosting gives the highest accuracy of 89%, MCC of 79.3% and Brier score of 0.075. On the other hand, the lowest result was given by Naive Bayes with 77% Accuracy, MCC of 54.6% and G-mean of 77.1%.

Table 1: Overall Result without feature selection.

Model	Accuracy	F1 Score	MCC	Brier Score	G-mean	H-Measure
LR	0.79	0.78	0.572	0.149	0.784	0.7859037497
DT	0.84	0.84	0.673	0.164	0.836	0.8363230211
RF	0.89	0.89	0.773	0.076	0.886	0.8865391896
KNN	0.88	0.88	0.762	0.096	0.873	0.8809876873
NB	0.77	0.77	0.546	0.186	0.771	0.7728909471
SVM	0.85	0.85	0.693	0.12	0.845	0.846558727
GB	0.89	0.9	0.793	0.075	0.896	0.8964263171
XGB	0.89	0.89	0.775	0.083	0.887	0.8877038821

Both Random Forest and XGBoost produced the best result of 88% after applying Information Gain; however, XGboost stood out as the best model between Random Forest and XGBoost for this feature selection technique, with 76.5% MCC and 88.2% G-mean value. After that, we employed Kendall's Tau feature selection method, in which gradient boosting outperforms the other methods with 89% accuracy and F1 score, 88.5% G-mean, 77.1% MCC, and 0.08 Brier score. Thirdly, we applied the Fisher score, where the highest accuracy of 89%, MCC of 77.6%, and H-measure of 88.8% are obtained by Gradient Boosting. Finally, we used Lasso L1, which gave us the best overall project result. We have selected top 7 features (age, bmi, avg_glucose_level, work_type, heart_disease, ever_married, and hypertension) in this feature selection technique to apply models. The Lasso L1 method, for feature selection was identified as the effective with the Gradient Boosting (GB) model achieving an accuracy of 90%, F1 score of 90%, Matthew's correlation coefficient (MCC) of 79.8%, Brier score of 0.076, G mean of 89.9% and an H measure of 89.9%.

Table 2: Result with feature selections.

Feature Selection	Model	Accuracy	F1 Score	MCC	Brier Score	G-mean	H-Measure
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Information Gain	LR	0.79	0.79	0.583	0.148	0.79	0.454
	DT	0.83	0.83	0.665	0.168	0.831	0.49
	RF	0.88	0.88	0.759	0.089	0.879	0.683
	KNN	0.83	0.83	0.661	0.126	0.825	0.541
	NB	0.78	0.78	0.564	0.167	0.782	0.455
	SVM	0.8	0.8	0.606	0.146	0.796	0.464
	GB	0.84	0.84	0.683	0.118	0.839	0.569
	XGB	0.88	0.88	0.765	0.087	0.882	0.683
Feature Selection	Model	Accuracy	F1 Score	MCC	Brier Score	G-mean	H-Measure
Kendall's Tau	LR	0.79	0.79	0.576	0.148	0.786	0.79
	DT	0.82	0.82	0.65	0.163	0.824	0.82
	RF	0.84	0.84	0.68	0.11	0.838	0.84
	KNN	0.82	0.82	0.655	0.127	0.819	0.83
	NB	0.77	0.77	0.537	0.186	0.767	0.77
	SVM	0.8	0.8	0.608	0.145	0.798	0.80
	GB	0.89	0.89	0.771	0.083	0.885	0.89
	XGB	0.85	0.85	0.694	0.106	0.846	0.85
Feature Selection	Model	Accuracy	F1 Score	MCC	Brier Score	G-mean	H-Measure
Fisher Score	LR	0.78	0.78	0.568	0.149	0.782	0.78
	DT	0.82	0.82	0.641	0.18	0.808	0.82
	RF	0.86	0.86	0.717	0.094	0.857	0.86
	KNN	0.84	0.84	0.696	0.124	0.841	0.85
	NB	0.77	0.77	0.533	0.188	0.766	0.77
	SVM	0.81	0.81	0.624	0.141	0.806	0.81
	GB	0.89	0.89	0.776	0.079	0.888	0.89
	XGB	0.87	0.87	0.74	0.092	0.869	0.87
Feature Selection	Model	Accuracy	F1 Score	MCC	Brier Score	G-mean	H-Measure
Lasso L1	LR	0.79	0.78	0.572	0.149	0.784	0.79
	DT	0.82	0.82	0.643	0.178	0.822	0.82
	RF	0.86	0.86	0.725	0.094	0.861	0.86
	KNN	0.84	0.83	0.681	0.12	0.829	0.84
	NB	0.77	0.77	0.54	0.185	0.769	0.77
	SVM	0.81	0.81	0.622	0.139	0.806	0.81
	GB	0.90	0.90	0.798	0.076	0.899	0.90
	XGB	0.87	0.87	0.745	0.09	0.872	0.87

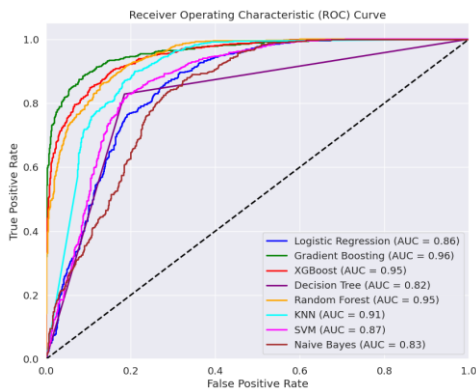


Figure 4: ROC curve of all machine learning models.

The resulting ROC curve plot provides a visual representation of the performance of each classifier in terms of the true positive rate vs the false positive rate. Here we can see Gradient Boosting model can perfectly separate the positive and negative classes, achieving the highest area under the curve value of 96% (shown in Figure 4).

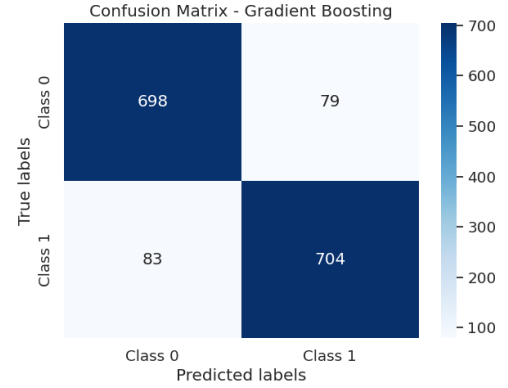


Figure 5: Confusion Matrix of Gradient Boosting (Lasso L1).

5 Conclusion & Future Work

Stroke has become a life-threatening condition in today's society. It is crucial to prevent strokes by detecting them as early as possible. Machine learning can play a key role in early stroke detection. We employed various machine learning algorithms with and without feature selection techniques in our research. Our findings indicate that the Lasso L1 feature selection technique is the most effective for Stroke Prediction. According to this technique, features such as age, bmi, avg_glucose_level, work_type, heart_disease, ever_married, and hypertension are crucial for stroke prediction. The Gradient Boosting model can accurately distinguish between positive and negative classes, achieving an impressive 96% area under the curve value. Our next work will involve incorporating additional stroke datasets from hospitals. Additionally, we plan to implement explainable AI techniques such as Permutation Importance, LIME, and Morris Sensitivity Analysis to gain a better understanding of the model's decision-making process both globally and locally. In the future, we will explore the implementation of AutoML (Automated Machine Learning) to automate data preprocessing, feature engineering, and model selection.

Acknowledgements

We, the authors, would like to express our sincere appreciation to the Bangladeshi Hospital (We left out the name to maintain anonymity) for their unconditional support while collecting data from stroke patients. This research work wouldn't have been possible without their invaluable help. Once again, we truly appreciate the administrative persons and doctors for playing a pivotal role in this research.

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